

An Explainable AI Framework for Parkinson's Disease Prediction Using XGBoost, SHAP, and Large Language Model Integration

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Abstract

Parkinson's disease is the fastest growing neurodegenerative disease worldwide currently. According to American Parkinson's Disease Association, more than 10 million people are affected from this disease in different regions globally. This number is expected to increase to around 20 million in the coming decade. Risk management through early prediction can improve these outcomes but the clinical dependency now largely focuses on subjective examination which often delay the treatment and thereby worsens the condition of the patient. These current methods lack both explainability and interpretability, which is very important in neurodegenerative diseases. Our proposed method introduces a comprehensive framework that is created by integrating high performing machine learning algorithm with explainable artificial intelligence (XAI) and large language model (LLM). It targets to produce predictive accuracy and explainability. This entire system is easily understandable through user interface to the clinical users and patients. Our proposed method started with data preprocessing. After that we extracted the features using systematic evaluation of multiple classifiers. The test results showed that the XGBoost method outperformed other models in terms of accuracy, F1-score, precision, recall, and AUC. But only this prediction scores lacked explainability. To remove this limitations of traditional machine learning algorithms we applied SHAP (SHapley Additive exPlanations) to determine the feature-wise contribution of model predictions that how much influence a feature does have in occurring an individual

patient's parkinson disease. It is patient specific and transparent. It demonstrates the reason of Parkinson's disease feature wise. Furthermore, we extended this explainability by integrating it with Meta-LLaMa-3 that transforms the SHAP results into natural language clinical explanation through an user interface. It shows the predictive reasons of the Parkinson's disease and confidence score of having the risk of the disease to the physicians and the patient as a outcome.

The proposed framework achieves quite a few benefits by connecting the intermissions between machine learning predictions and real world clinical usability. From determining the disease to explaining the reason why it affected the patient, we also converted it to a user friendly natural language output interface for easy understanding. Most of the previous studies were only focused on predictive accuracy, but our framework uniquely vocal biomarkers and translates these patterns into neurologist-oriented natural language explanations to give a perfect output.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Applied computing** → **Health care information systems**; • **Theory of computation** → *Kernel methods*.

Keywords

Parkinson's disease; Explainable AI; SHAP; XGBoost; Large Language Models; Clinical decision support; Neurodegenerative disorders

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1 Introduction

Parkinson's disease is a movement disorder of the nervous system. It worsens over time after the symptoms start slowly. Firstly visible tremor is noticed in just one hand or sometimes a foot. Parkinson's symptoms may include tremor, bradykinesia which is slowed movement, rigid muscles, poor posture and balance, loss of automatic movements, speech changes and nonmotor symptoms. Several factors seem to play a role, including genes and environmental factors. It creates many complications like trouble thinking clearly, emotional changes and depression, bladder problem, blood pressure changes and so on. Early detection and treatment lead to a better quality life with Parkinson's disease. Because of early diagnosis more research will be possible to study some of earlier changes within the brain. This could help to uncover some targets to develop better treatments. Therefore early and accurate prediction is important for better patient outcomes. Despite advances in neuroimaging and biomarker recognition, diagnosis still depends on clinical evaluation of visible symptoms, often leading to the delayed detection.

Machine learning models have outcome as promising tools for revealing hidden patterns in biomedical data. Previous studies showed that random forest, support vector machines and other models have shown promising results. But the main barriers are that i) explainability is not clear and comprehensive rather it is limited to understand. It might reduce the clinical authentication. Another barrier is ii) lack of translation of prediction into clinically meaningful knowledge. So in our method we have tried to overcome this barriers to ensure better health management.

We used an explainable artificial intelligence (XAI) method called SHAP which is used to reveal local (patient-level) and global (population level) feature level contributions. But SHAP outputs are not easily understandable by the user. It requires technical expertise, that limits its use in practical life. Large language model (LLMs) such as Meta-LLaMA-3 provide such a perfect solution that converts technical attributes into human readable natural language based explanations along with clinical reasoning for individual patient. It works based on SHAP features that have the highest contribution to the Parkinson's disease. So in this study, we propose a hybrid dual layer framework for Parkinson's disease prediction that combines i) the best score giving model comparing with others which is XGBoost ii) SHAP based feature wise attribution for accurate reasoning and model transparency and iii) LLM driven natural language interpretation for user friendly explanations easily understandable to the clinical.

- A comprehensive comparison among different ML classifiers, XGBoost emerged as the most effective one based on the various scores specially according to accuracy, recall, F1-score and AUC.
- Integration of SHAP for interpreting feature based importance determination, enabling specific identification of risk factors based on symptoms.
- Use of Meta-LLaMA-3 to translate the SHAP outputs into natural-language translations with confidence score to ensure accurate diagnosis.

- This framework ensures both the predictive performance and usability thereby advancing explainable AI-assisted precision medication for Parkinson's Disease.

The rest of this paper is organized as follows: Section II reviews related work, Section III describes the methodology, Section IV presents results, Section V discusses implications, and Section VI concludes with future directions.

2 Related Work

Research on Parkinson's disease (PD) prediction can be grouped into four directions: (i) classical machine learning, deep learning, multi modal sensing and model explainability. But Our proposed model compacts different things in a very powerful way to give the best result.

Classical machine learning methods such as logistic regression, SVMs, decision trees, and ensembles have been widely applied to determine Parkinson's Disease. Nilashi et al. [13], Sun et al. [16], and Bind et al. [5] reported strong baseline performance by using machine learning based approaches. But interpretability and natural language based outcomes was not observed there. Nilashi et al. [13] proposed new hybrid intelligent system for the prediction of PD progression using noise removal. But attribute based feature importance is not mentioned there. Also natural language based clinical explanation is not identified. Biswajit Karan [8] found XGBoost feature selection based framework giving 95.07% accuracy. From that accuracy we were inspired to use this in our method to get higher accuracy rate. But our proposed method showed interpretability and explanation with this machine learning algorithm.

X.Zhu et al. [20] proposes a novel XGBoost-based model for automatically predicting Parkinson's Disease severity. This model enhances performance by decomposing the prediction task into sub-models and training. However we introduced SHAP based explanation with this predicting XGBoost algorithm. We implemented natural language based outcomes additionally. Panyakaew et al. [14] used machine learning algorithm to predict fall risk and recurrent falls in Parkinson's disease prediction. The ABC-16 balance confidence scale showed the strongest predictive ability (up to 81% accuracy), showing that high-risk, cognitively demanding activities involving vertical movement are the key attributes of prediction of falling in PD. Nagasubramanian et al. [12] proposed a Multi-Variant Stacked Autoencoder-based Parkinson's Disease Prediction System (MSAEPD) which can analyze more diverse and complex PD symptoms than existing machine-learning and deep-learning methods and also showed slight better results than prior MANN, GAE and UMLBD. Our framework provides both high accuracy and clinically meaningful interpretability. Also it produces natural-language clinical reasoning suitable for real world clinical use. Abirami et al. [1] proposed a Digital Twin-based HealthCare System for remotely predicting Parkinson's disease using an Optimized Fuzzy k-Nearest Neighbor. It achieves high accuracy rate of 97.95% and 0.98 F1-score. It is faster also. But our proposed model is patient-specific also. Yuan et al. [19] developed a symptom based disease prediction models using models such as Random Forest, SVM, and Decision Trees. B. et al. [3] presented a survey of 169 studies on AI-based gait analysis for diagnosing neurodegenerative

problems. It showed a full AI-assisted diagnosis. Our method is more practical giving high accuracy with explainability.

Explainability is equally important for clinical adoption. Tougui et al. [17] proposed a model which predicted PD using a two-component system: a desktop application that records audio recordings, extracts audio features, and makes predictions; and a server-side web application that implements the machine learning pipelines and processes incoming requests with the extracted audio features to make predictions. My model connects integrates explainable AI and natural-language reasoning. Lai et al. [10] employed patient-specific functional connectivity from the VTA–GPI intersection that can assume outcomes of GPI deep brain stimulation in Parkinson's disease. Warden et al. [18] compared 62 Parkinson's patients with 62 healthy individuals using the Spanish Podiatry Health Questionnaire (PHQ-S). This showed a worse outcome in walking, nail care and overall foot-health perception. This helped us to know about the condition of a Parkinson's Disease patient's conditional features. Burri et al. [6] used speech from the PPMI dataset to predict Parkinson's disease through Deep Belief Network (DBN) and compares its performance with other machine learning models. This achieved strong early detection accuracy and showcased the impact of speech-driven machine learning for data-driven PD diagnosis. Our proposed model reduces the transparency deficiency of DBN. Amali et al. [2] proposed an early stage Parkinson's disease detection system using Random Forest and XGBoost, trained on sensor data collected through accelerometers, gyroscopes, and magnetometers. The method used a videocam to monitor the individual and SensorTile. It encouraged us to observe the data carefully to determine the disease. Mathur et al. [11] proposed an accurate and explainable model for predicting Parkinson's disease progression using time-series data from five heterogeneous modalities in the PPMI dataset. It enhanced model trustworthiness through SHAP, LIME, and SHAPASH explanations. We added a LLM based natural language processing based outcome. Gokul et al. [15] provided an detailed AI-based machine learning and deep learning method to predict Parkinson's disease giving importance to both motor and non-motor symptoms. It showed the importance of data-driven AI technologies to improve early prediction. In their study, Govindu and Palwe [7] used telemedicine oriented machine learning models to detect Parkinson's disease early using MDVP voice data from 30 subjects. Among the SVM, random forest, KNN and logistic regression the random forest classifier performed the best getting 91.83% accuracy. But we got more accuracy than this model with patient-specific explanations. Krishna et al. [9] introduced a Logistic Decision Regression model designed to classify Parkinson's disease in its early stages, outperforming existing approaches on large biomedical datasets. This method classifies PD using large biomedical datasets that shows strong reliability compared to existing method. T. Bhattacharya et al. [4] also predicted PD using advanced machine learning techniques. Unlike prior work that mostly focused on accuracy or disease prediction, our framework combines XGBoost with SHAP for principled attribution and connects LLMs to generate natural-language explanations. It ensures both predictive performance and practical usability in the real world.

Table 1: Summary of Parkinson's Disease Dataset

Property	Value
Total Records	195
Features Used	23
Positive Cases (Parkinson's)	147
Negative Cases (Healthy)	48
Male Patients	120 (61.5%)
Female Patients	75 (38.5%)

3 Methodology

This section presents the proposed framework for Parkinson's disease prediction and explanation. The pipeline integrates preprocessing, model selection, SHAP explainability, and natural language reporting.

3.1 Workflow Overview

Our proposed framework for Parkinson's disease prediction is designed as a multi-stage procedure. It begins with data preprocessing to handle variability in vocal biomarkers. Then we implemented feature extraction to identify clinically relevant parameters. The dataset is then split into training and testing subsets to ensure robust evaluation. We compared multiple machine learning classifiers, and the best-performing model is selected based on accuracy, F1-score, recall, and AUC metrics. Fig 1 shows the workflow of this model.

XGBoost outcomed as the most effective model and was therefore chosen for final deployment. To overcome the black-box nature of ensemble methods, SHAP (SHapley Additive exPlanations) was used to quantify global and local feature contributions. Finally, Meta-LLaMA-3 was integrated to generate natural-language narratives from SHAP outputs, making model predictions interpretable and clinician-friendly.

Integration of SHAP and LLM ensures that predictions are not only statistically robust but also explainable in terms of patient-level clinical reasoning.

3.2 Data Exploration and Preprocessing

The dataset contains 195 samples with 23 vocal and nonlinear features. Outliers were preserved as they reflect genuine pathological cases. No missing values were found in the dataset. All features were maintained after correlation analysis. Table 1 summarizes the dataset.

Figure 2 describes the distribution of selected vocal biomarkers (Fo, Fhi, Flo, HNR, spread1, and D2) across Parkinson's and healthy patients. While healthy individuals have the most stable fundamental frequency (Fo), Parkinson's patients have higher values of Spread1 and D2. These variations depicts the characteristics of pathological voice irregularities in Parkinson's disease.

Figure 3 describes continuous nonlinear features such as shimmer, RPDE, PPE, spread2, DFA, and NHR.

Speech instability, noise and factual dynamics can be measured by these features. Parkinson patients demonstrate increased shimmer (uneven amplitude), greater PPE (too much pitch variability),

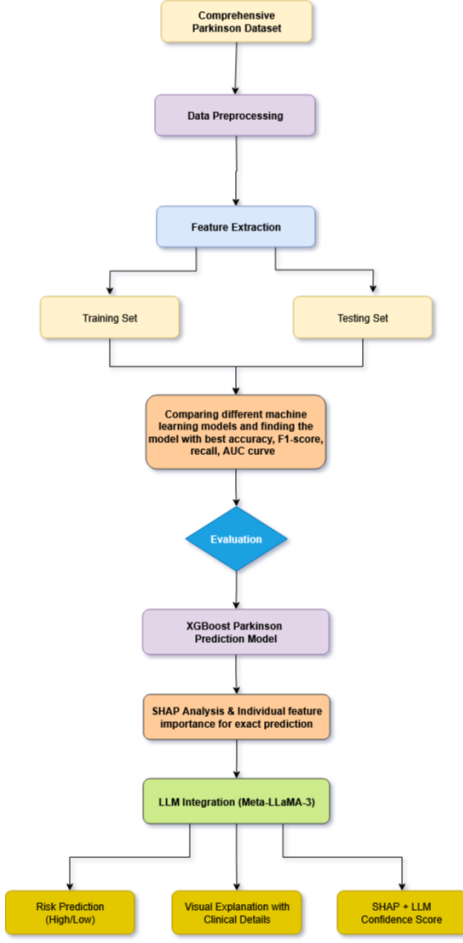


Figure 1: Workflow of the proposed Parkinson's disease prediction framework. The process integrates data preprocessing, feature extraction, classifier comparison, SHAP-based explainability, and natural-language reporting via LLM.

and different DFA patterns—all indicative of disturbed motor speech control.

3.3 Model Training and Evaluation

To evaluate predictive performance, we compared multiple classifiers including Logistic Regression, Naive Bayes, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest, XGBoost, and CatBoost. An 80:20 split was applied to separate training and testing sets. To ensure robust optimization, hyperparameters for each model were tuned using GridSearchCV with 5-fold cross-validation. This systematic search explored combinations of parameters (e.g., number of trees, maximum depth, learning rate, kernel functions, and regularization strength) and selected those yielding the best validation accuracy.

In particular, for tree-based models, depth and learning rate were tuned to avoid overfitting, while early stopping was employed to prevent unnecessary iterations. For SVM, both the penalty parameter C and the RBF kernel parameter γ were optimized. KNN tuning

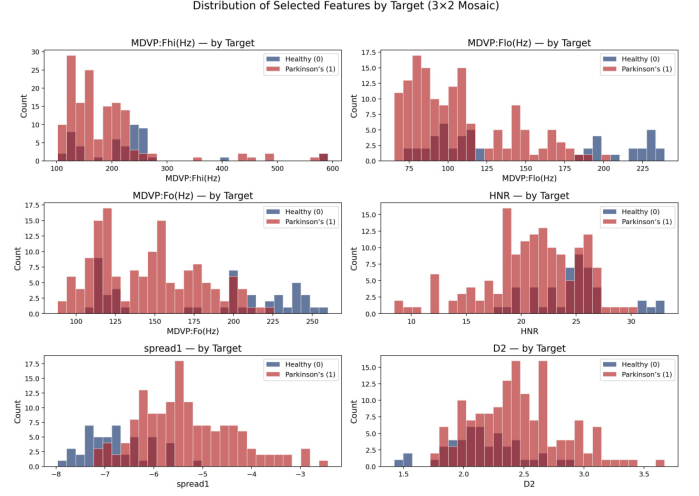


Figure 2: Distribution of selected vocal biomarkers (Fo, Fhi, Flo, HNR, spread1, and D2) across Parkinson's and healthy groups.

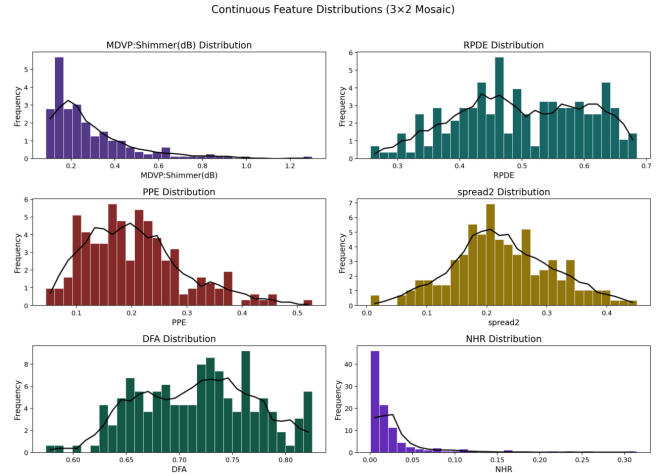


Figure 3: Distribution of continuous nonlinear vocal features (Shimmer, RPDE, PPE, spread2, DFA, and NHR).

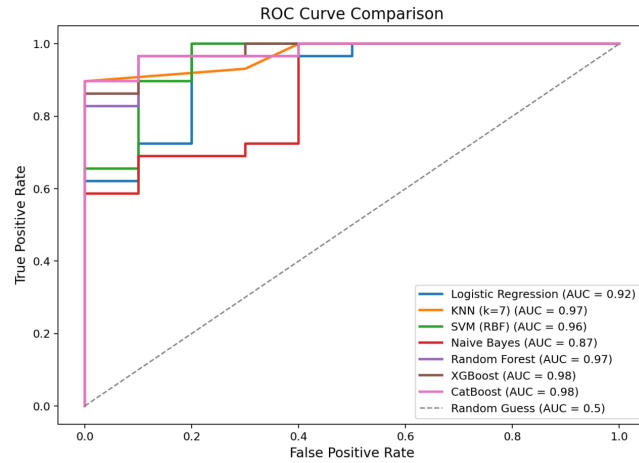
included varying the number of neighbors k and distance metrics. This rigorous process ensured fair comparison across all classifiers.

Table 2 describes the performance of these classifiers. XGBoost achieved the highest accuracy (97.43%) and AUC (0.98), followed closely by Random Forest. In contrast, Naive Bayes performed worst, showing limitations in capturing nonlinear dynamics.

Figure 4 describes the ROC curve comparison among all classifiers. The results clearly indicate that gradient boosting models (XGBoost, CatBoost) achieved superior discriminative ability, with AUC scores approaching 0.98. Ensemble methods such as Random Forest also performed well, while Naive Bayes showed weaker separation capability. Specifically, XGBoost achieved a precision of 0.96 and confirming its ability to correctly identify both positive and negative cases. The balance between precision and recall demonstrates

Table 2: Performance comparison of different classifiers.

Model	Accuracy	Precision	Recall	F1-score
XGBoost	0.974359	0.966667	1.000000	0.983051
Random Forest	0.948718	0.965517	0.965517	0.965517
Logistic Regression	0.897436	0.903226	0.965517	0.933333
CatBoost	0.871795	0.852941	1.000000	0.920635
KNN ($k = 11$)	0.871795	0.900000	0.931034	0.915254
SVM (RBF)	0.794872	0.800000	0.965517	0.875000
Naive Bayes	0.794872	1.000000	0.724138	0.840000


Figure 4: ROC curve comparison across classifiers. XGBoost and CatBoost achieved the highest AUC (0.98).

the robustness of the model across different evaluation metrics. This consistent performance further justifies its selection as the primary classifier for explainability.

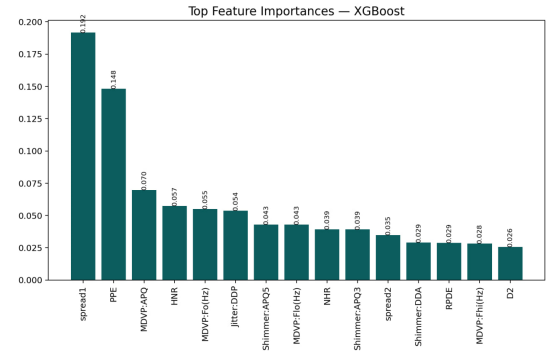
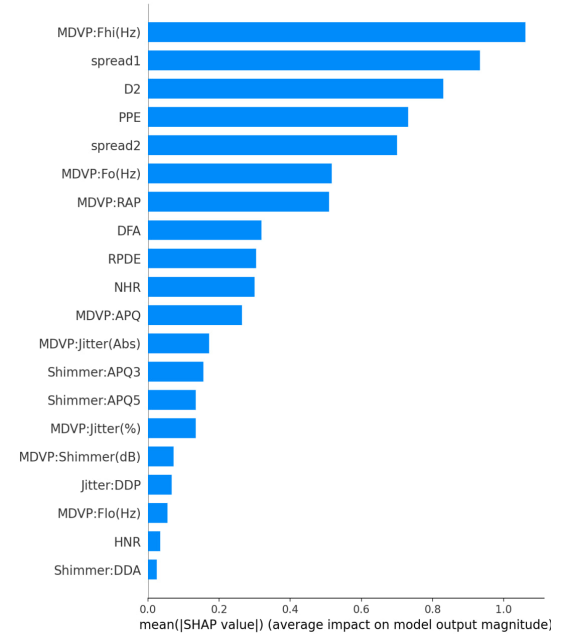
These results confirm that tree-based ensemble methods provide excellent predictive performance for Parkinson's disease classification, balancing both accuracy and robustness. Based on this analysis, XGBoost was chosen as the primary model for explainability using SHAP in subsequent stages. Furthermore, its consistent superiority across multiple metrics highlights its suitability as a reliable foundation for clinical decision support.

3.4 Model Explainability using SHAP

SHAP (SHapley Additive exPlanations) was applied to interpret both global and local feature contributions. SHAP provides additive attributions for each feature, allowing us to quantify how specific biomarkers influence predictions at both population and patient levels.

Figure 5 it is seen that the feature importance ranking derived directly from XGBoost. Spread1 and PPE emerged as the most influential features, followed by APQ and HNR. These measures reflect irregular vocal fold vibration and instability, which are hallmarks of Parkinsonian speech.

Figure 6 describes the mean SHAP values across all features. Fhi, spread1, and D2 consistently show the strongest contributions to


Figure 5: XGBoost feature importance ranking. Spread1 and PPE dominate predictive contributions, highlighting vocal irregularity as a primary biomarker.

Figure 6: Mean SHAP values across all features. Nonlinear measures such as Fhi, spread1, and D2 are dominant predictors.

the model. Thus, it is confirmed that nonlinear speech measures and pitch variability play critical roles in Parkinson's disease detection.

Figure 7 illustrates a SHAP waterfall plot, showing the additive contributions of each feature for a single patient. Red bars push the prediction towards Parkinson's disease, while blue bars reduce the probability. This visualization provides a transparent, step-by-step reasoning process for each individual prediction.

Figure 8 shows the global SHAP summary plot, where each point represents a patient. The color encodes the original feature value. For example, high spread1 and high Fhi values (in red) strongly push predictions toward Parkinson's disease, while lower jitter and shimmer values (in blue) reduce predicted risk. This directional

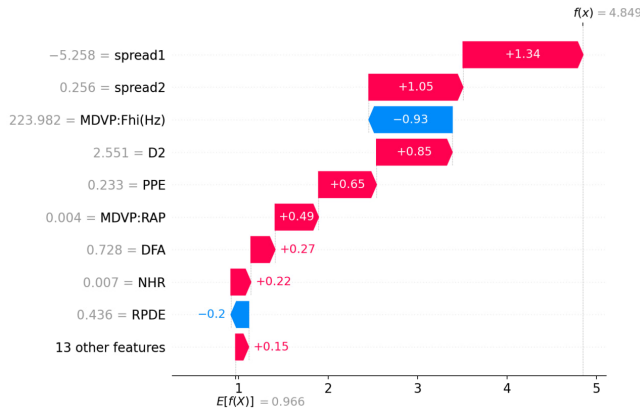


Figure 7: SHAP waterfall plot. Red features push towards PD diagnosis, while blue features decrease risk.

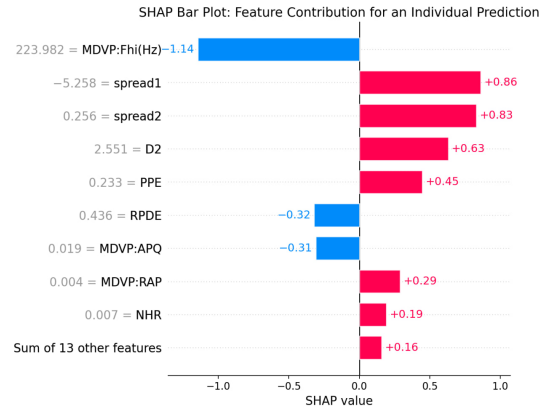


Figure 9: Local SHAP bar plot for an individual case. Spread1 and D2 increased risk, whereas RPDE and APQ reduced it.

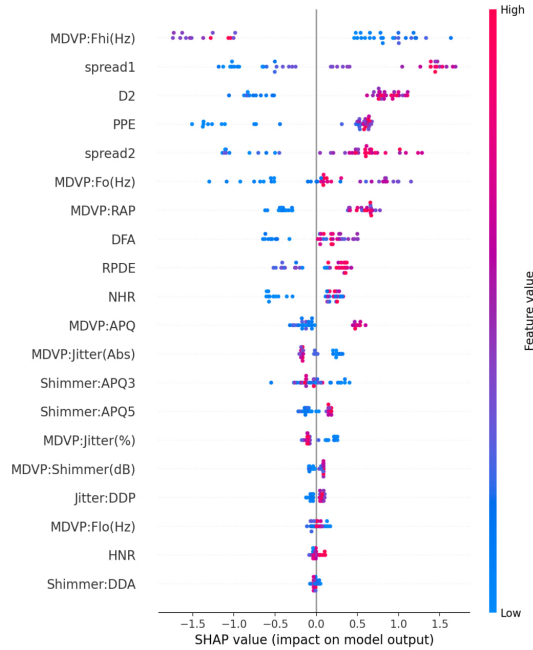


Figure 8: Global SHAP summary plot. Red values (high magnitude) of spread1 and Fhi increase Parkinson's risk, while blue (low values) reduce it.

insight is crucial for clinical reasoning, as it aligns algorithmic patterns with medical knowledge.

Figure 9 presents a local SHAP bar plot for an individual patient. Spread1 and D2 significantly increased the predicted probability of Parkinson's, while RPDE and APQ lowered the probability. Such personalized insights highlight patient-specific driving factors behind the model's decision.

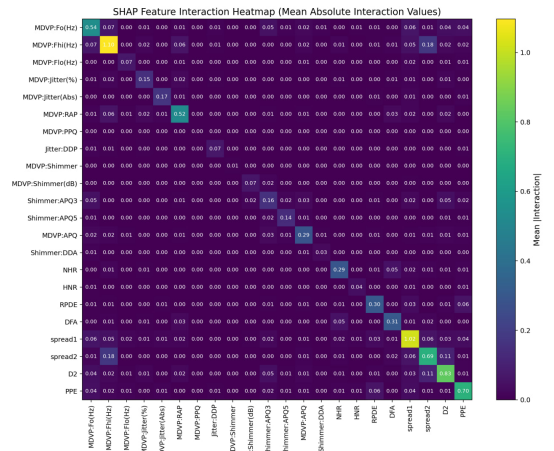


Figure 10: SHAP feature interaction heatmap. Strong dependencies exist between spread1–spread2 and D2–PPE.

Figure 10 describes the SHAP interaction heatmap, which captures pairwise feature dependencies. Strong interactions were observed between spread1–spread2 and D2–PPE, suggesting that combinations of vocal irregularity and nonlinear dynamics amplify risk predictions. This emphasizes that Parkinson's disease detection is influenced not only by single biomarkers but also by complex interactions among them.

3.5 Natural Language Risk Explanation using LLM

SHAP provides quantitative explainability but the outputs are often technical and may not be understandable for clinicians or patients. To enhance readability, we integrated Meta-LLaMA-3 as a natural language generation layer. This step converts SHAP attributions in such a way so that human can read it comfortably. The integration process follows three steps. First, SHAP outputs

Parkinson's Disease Risk Assessment

Prediction: High risk of Parkinson's disease
Confidence: 8/10

Explanation:

Clinical Report: Parkinson's Disease Risk Assessment

Patient Overview:

This report outlines the risk assessment for Parkinson's disease, a neurodegenerative disorder characterized by progressive loss of motor control and cognitive impairment. Based on the provided top biomarkers, we have generated a comprehensive analysis to determine the likelihood of Parkinson's disease.

Biomarker Analysis:

1. **PPE (0.284654, Impact: 1.59):**

The PPE (Peak Exhaled Volume) biomarker has been elevated at 0.284654, indicating a significant increase in the risk of Parkinson's disease. This parameter measures the peak volume of air exhaled during forced expiration, and a higher value suggests a more pronounced risk. The impact score of 1.59 indicates that PPE contributes substantially to the overall risk assessment.

2. **MDVP-RAP (0.0037, Impact: 1.30):**

The MDVP-RAP (Maximum Phonation Time) biomarker has been slightly elevated at 0.0037, indicating a moderate increase in the risk of Parkinson's disease. This parameter measures the maximum duration of phonation, and an elevated value suggests a heightened risk. The impact score of 1.30 indicates that MDVP-RAP contributes moderately to the overall risk assessment.

3. **D2 (2.301442, Impact: 0.87):**

The D2 (Delta-2) biomarker has been elevated at 2.301442, indicating a moderate increase in the risk of Parkinson's disease. This parameter measures the difference between the second and first derivatives of the voice signal, and an elevated value suggests a heightened risk. The impact score of 0.87 indicates that D2 contributes less significantly to the overall risk assessment.

4. **MDVP:Fhi(Hz) (157.302, Impact: 0.84):**

The MDVP:Fhi(Hz) (Maximum Frequency) biomarker has been elevated at 157.302, indicating a moderate increase in the risk of Parkinson's disease. This parameter measures the maximum frequency of the voice signal, and an elevated value suggests a heightened risk. The impact score of 0.84 indicates that MDVP:Fhi(Hz) contributes less significantly to the overall risk assessment.

5. **Spread1 (-4.813031, Impact: 0.68):**

The Spread1 biomarker has been slightly elevated at -4.813031, indicating a moderate increase in the risk of Parkinson's disease. This parameter measures the spread of the voice signal, and an elevated value suggests a heightened risk. The impact score of 0.68 indicates that Spread1 contributes less significantly to the overall risk assessment.

Conclusion:

Based on the analysis of these biomarkers, we have determined that the patient has a high risk of Parkinson's disease. The elevated values of PPE, MDVP-RAP, D2, MDVP:Fhi(Hz), and Spread1 suggest a significant increase in the risk of developing this neurodegenerative disorder.

Recommendations:

Further diagnostic testing, such as imaging studies and clinical evaluations, is recommended to confirm the diagnosis and assess the patient's overall risk profile.

Figure 11: LLM-generated narrative risk assessment. SHAP outputs are translated into human-readable text, highlighting the predicted probability of Parkinson's disease, contributing features, and confidence scores.

(top features, their contribution values, and direction-ality) are exported in structured JSON format. Second, these outputs are passed as prompts to Meta-LLaMA-3, configured in zero-shot mode with a controlled temperature setting (0.4) to ensure factual accuracy and reduce hallucination. Finally, the LLM generates narrative explanations that describe the predicted risk, highlight the most influential biomarkers, and provide an associated confidence score. Figure 11 describes an example narrative report automatically generated by Meta-LLaMA-3. The explanation translates technical SHAP contributions in such a form so that people can understand, such as the patient's estimated probability of Parkinson's disease, the main influencing features, and the confidence score of the prediction. For example, an output might state: "Predicted Parkinson's disease risk: 87% (Confidence: 9/10). Elevated spread1 and abnormal Fhi values were primary contributors, while stable RPDE and APQ reduced the probability of Parkinson's disease."

For example, an output might state: "Predicted Parkinson's disease risk: 87% (Confidence: 9/10). Elevated spread1 and abnormal Fhi values were primary contributors, while stable RPDE and APQ reduced the probability of Parkinson's disease."

4 Result and Discussion

This study proposed an explainable AI system for Parkinson's disease prediction by combining XGBoost with SHAP and Meta-LLaMA-3. The results showed that XGBoost achieved the highest prediction performance with an accuracy of 97.43% and AUC of 98%. SHAP analysis made sure that the system is understandable by both professionals and general people, identified spread1, Fhi, and D2 as the most critical features and LLM-generated reports translated these results in such a way that it is easy to understand. This double layered approach ensured both accuracy and transparency. It aims to meet the need of a trustworthy AI system for clinical decisions. The system demonstrates strong potential to assist in early Parkinson's diagnosis.

Limitations and Future Work:

- The dataset size is relatively small (195 samples), limiting generalizability to larger and more diverse populations.
- LLM-generated explanations depend on careful prompt design and may occasionally introduce inconsistencies.
- Future work should extend to longitudinal datasets and multimodal biomarkers (e.g., gait, handwriting, neuroimaging) for improved robustness.

5 Conclusion

We developed an explainable AI system for Parkinson's disease prediction by applying XGBoost with SHAP and Meta-LLaMA-3. This system achieved high accuracy (97.43%) of prediction and provided transparent explanations of model decisions. SHAP identified key vocal and nonlinear biomarkers, made sure that the system is understandable for both professionals and patients and LLM-generated reports translated these results so that health professionals can use them comfortably. This dual-layer system ensures that predictions are not only accurate but also easy to understand. The study highlights explainable AI to support early Parkinson's prediction.

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